5. What connection is used in domestic appliances and why?

VIII. Long answer questions.

1. With the help of a circuit diagram derive the formula for the resultant resistance of three resistances connected: a) in series and b) in parallel

2. a) What is meant by electric current?
   b) Name and define its unit.
   c) Which instrument is used to measure the electric current? How should it be connected in a circuit?

3. a) State Joule's law of heating.
   b) An alloy of nickel and chromium is used as the heating element. Why?
   c) How does a fuse wire protect electrical appliances?

4. Explain about domestic electric circuits. (circuit diagram not required)

5. a) What are the advantages of LED TV over the normal TV?
   b) List the merits of LED bulb.

IX. Numerical problems:

1. An electric iron consumes energy at the rate of 420 W when heating is at the maximum rate and 180 W when heating is at the minimum rate. The applied voltage is 220 V. What is the current in each case?

2. A 100 watt electric bulb is used for 5 hours daily and four 60 watt bulbs are used for 5 hours daily. Calculate the energy consumed (in kWh) in the month of January.

3. A torch bulb is rated at 3 V and 600 mA. Calculate it's
   a) power
   b) resistance
   c) energy consumed if it is used for 4 hour.

4 A piece of wire having a resistance R is cut into five equal parts.
   a) How will the resistance of each part of the wire change compared with the original resistance?
   b) If the five parts of the wire are placed in parallel, how will the resistance of the combination change?
   c) What will be ratio of the effective resistance in series connection to that of the parallel connection?

X. HOTS:

1. Two resistors when connected in parallel give the resultant resistance of 2 ohm; but when connected in series the effective resistance becomes 9 ohm. Calculate the value of each resistance.

2. How many electrons are passing per second in a circuit in which there is a current of 5 A?

3. A piece of wire of resistance 10 ohm is drawn out so that its length is increased to three times its original length. Calculate the new resistance.

REFERENCE BOOKS

1. Electrodynamics by Griffiths
2. Fundamentals of Electric Circuits by Charles Alexander

INTERNET RESOURCES

https://www.elprocus.com/basic-electrical-circuits-and-their-working-for-electrical-engineers/

https://www.physicsclassroom.com/calcpad/circuits

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